CHAPTER IV

RESULTS AND DISCUSSIONS

The introduction provides a brief overview of what is presented in Chapter IV. It explains that the chapter presents, analyzes, and discusses the results obtained from the system design, development, testing, and evaluation. The discussion should be organized according to the specific research objectives to ensure that each objective is addressed by corresponding results and interpretations.

I. Results of System Design and Development Per Module

*This section presents the results of the **design and development phase of the system**. It should demonstrate how the identified user requirements, functional requirements, and technical requirements were translated into the proposed system architecture, interfaces, database, processes, and other design components.*

System Architecture

This section presents the overall technical structure of the developed system. It shows how the different components of the system interact with one another, including users, applications, databases, servers, networks, APIs, or other technologies used in the study.

The system architecture should help the reader understand how information flows through the system from input to processing and output.
What to include:

- *Present the System Architecture Diagram.*
- *Identify the major components.*
- *Explain the role of each component.*
- *Describe how the components communicate.*
- *Explain the flow of data.*
- *Explain how users' access and interact with the system.*
- *Identify the database or storage component, if applicable.*

Screenshots of System Design and Development Per Module

This section presents the screenshots of the developed system interfaces for each module. For every module, the researchers should explain what the module is designed to accomplish; explain how the module was developed based on the requirements and system design; describe the major functions that users can

perform; explain how users use the interface to perform a specific task; and describe what happens after the user performs the required action.

Results of Unit Testing Per Module

This section presents the results of unit testing, which determines whether individual system components, functions, or modules operate correctly when tested independently by the researchers. The purpose is to demonstrate that each individual component performs according to its intended function before the components are integrated.

Table #. Results of Unit Testing for Module XYZ (Sample)

Module	Function Tested	Expected Result	Actual Result	Status
Login	*User authentication*	*Valid user is allowed to log in*	*User successfully logged in*	*Passed*
Registration	*Account creation*	*User information is saved*	*Information successfully saved*	*Passed*
User Management	*Update user information*	*Information is updated*	*Information successfully updated*	*Passed*
Report	*Generate report*	*Report is generated*	*Report successfully generated*	*Passed*

Note: Discuss the overall result.
The discussion should identify:

- *How many functions were tested.*
- *How many passed or failed.*
- *Whether errors were encountered.*
- *What corrections were made, if applicable.*
- *Whether the individual components performed according to their expected behavior.*

Results of Integration Testing

This section presents the results of integration testing, which determines whether the different modules or components of the system work correctly when combined. While unit testing checks individual components, integration testing checks whether those components can properly communicate, exchange data, and work together.

Table #. Results of Unit Testing for Module XYZ (Sample)

Integrated Modules	Test Scenario	Expected Result	Actual Result	Status
Registration + Database	*Register a new user*	*User information is saved in the database*	*Information successfully saved*	*Passed*
Login + User Management	*Log in using registered account*	*System authenticates the user*	*User successfully authenticated*	*Passed*
Transaction + Database	*Submit transaction*	*Transaction is stored and retrieved correctly*	*Transaction successfully stored*	*Passed*
Report + Database	*Generate report*	*Report displays the correct stored information*	*Correct information displayed*	*Passed*

Note. The discussion should explain whether the modules successfully interacted with one another.

It should answer:

- *Did the modules communicate correctly?*
- *Was data transferred correctly between modules?*
- *Were there errors during integration?*
- *Were identified errors corrected?*
- *Did the integrated functions produce the expected results?*

II. Performance Evaluation Results by the Stakeholders

This section presents the results of the evaluation of the developed system by the identified respondents or evaluators.

III. Performance Evaluation Results by the IT Experts

This section presents the results of the evaluation of the developed system by the identified respondents or evaluators.